Surface Water Quality Index Estimation via Gradient Boosted Decision Trees and Stoichiometric Feature Engineering

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Project 18: Surface Water Quality Prediction — Semester 5
Repository: https://github.com/MAYANKSHARMA01010/Semester_5_AML_DL

Abstract—Automated surface water quality monitoring is vital for real-time ecological conservation and public health safeguarding. However, conventional biological and chemical assays, such as the 5-day Biochemical Oxygen Demand (BOD_5), suffer from multi-day incubation latencies. This paper presents Phase 1 of an applied machine learning investigation predicting the Canadian Council of Ministers of the Environment Water Quality Index (CCME WQI) across a multi-national dataset of 2,827,977 sensor observations spanning 83 years (1940–2023) across five nations (England, USA, Ireland, China, and Canada). Following rigorous exploratory data auditing, 8,360 physically implausible sensor telemetry records were purged, yielding a complete-case ground truth of 2,819,617 records. A 38-dimensional feature representation was constructed incorporating non-linear $\log(1+x)$ transformations, aquatic stoichiometry (Redfield N:P ratio, Benson–Krause dissolved oxygen saturation, Emerson un-ionized ammonia equilibrium), and chronological station-level autoregressive lags. Benchmarked against a stratified 70/15/15 partition, Random Forest (R^2=0.9999, MAE=0.052) and XGBoost (R^2=0.9994, MAE=0.161, 9× faster inference) dramatically outperformed mean-baseline heuristics (MAE=14.130). Game-theoretic TreeSHAP attribution identified orthophosphate (ρ=−0.78 with WQI) and station memory as primary predictive drivers.

Index Terms—Water Quality Index, CCME WQI, Gradient Tree Boosting, Random Forest, Stoichiometric Feature Engineering, TreeSHAP, Applied Machine Learning.

I. Introduction

FRESHWATER ecosystems face unprecedented anthropogenic contamination from municipal wastewater, intensive agricultural runoff, and chemical effluents. Regulatory authorities worldwide monitor aquatic integrity through standardised physical and chemical parameters, including pH, dissolved oxygen (DO), biochemical oxygen demand (BOD), electrical conductivity, ammoniacal nitrogen, and orthophosphate [1].

To synthesise multi-dimensional telemetry into actionable policy directives, regulatory agencies rely on composite indicators, prominently the Canadian Council of Ministers of the Environment Water Quality Index (CCME WQI). Despite its diagnostic utility, computing the CCME WQI requires synchronous multi-variable laboratory testing; assays such as BOD_5 mandate five days of microbiological incubation, precluding real-time detection of acute pollution events.

Machine learning offers high-throughput surrogate inference, translating instantaneous sensor readings directly into continuous index scores. While tree-based ensembles demonstrate exceptional competence on tabular telemetry, naive deployments frequently encounter critical pitfalls: (i) sensor corruption, (ii) extreme right-skewed nutrient distributions, (iii) spatial-temporal data leakage, and (iv) uninterpretable decision boundaries.

This paper details the Phase 1 Machine Learning Evaluation for Project 18. Our contributions include:

1) Multi-National Dataset Audit: Auditing 2,827,977 raw records across five nations and purging 8,360 physical impossibility errors.
2) Domain Feature Engineering: Deriving 38 features grounded in aquatic chemistry (Henry's law, Redfield stoichiometry, Streeter–Phelps oxygen deficit, and non-lookahead station memory).
3) Rigorous AML Benchmarks: Comparing Random Forest and XGBoost with strict anti-leakage target quarantine.
4) Explainable AI (XAI): Employing game-theoretic TreeSHAP attribution to reveal environmental drivers.
5) Modular Code Quality: Delivering a production-grade Python package (src/) validated by 12 automated unit tests (pytest).

II. Related Work & Literature Review

Early computational water quality modelling relied on process-based differential systems, such as the classical Streeter–Phelps oxygen sag equations. While physically consistent, these numerical models require expensive bathymetric calibrations and fail to generalise across diverse hydrological basins.

With the proliferation of automated telemetry, data-driven approaches emerged. Breiman introduced Random Forests (RF), demonstrating that aggregating randomised decision trees drastically reduces variance while resisting

tabular noise. Chen and Guestrin developed XGBoost [2], utilising second-order gradient boosting, sparsity awareness, and regularised objective functions to achieve state-of-the-art accuracy on structured environmental datasets.

Lundberg and Lee unified cooperative game theory and tree ensembles through SHAP and TreeSHAP, enabling exact polynomial-time computation of Shapley values. Published water quality literature often derives index targets directly from raw features without preventing circular mathematical leakage, or overlooks physical thermodynamic constraints. Our Phase 1 pipeline explicitly bridges this gap.

III. Mathematical Formulation

A. CCME Water Quality Index

The CCME WQI integrates three mathematical dimensions into a scalar bounded in $[0,100]$:

$$\text{CCME WQI} = 100 - \left(\frac{\sqrt{F_1^2 + F_2^2 + F_3^2}}{1.732} \right) \tag{1}$$

where $1.732 = \sqrt{3}$ normalises the Euclidean distance from $(0,0,0)$ to worst-case $(100,100,100)$. The three components are:

Scope (F_1) — percentage of variables failing thresholds:

$$F_1 = \left(\frac{V_{\text{failed}}}{V_{\text{total}}} \right) \times 100 \tag{2}$$

Frequency (F_2) — percentage of individual tests failing:

$$F_2 = \left(\frac{N_{\text{failed}}}{N_{\text{total}}} \right) \times 100 \tag{3}$$

Amplitude (F_3) — normalised sum of excursions:

$$\text{excursion}_i = \begin{cases} \dfrac{\text{TestValue}_i}{\text{Guideline}_j} - 1 & \text{if failed upper} \\ \dfrac{\text{Guideline}_j}{\text{TestValue}_i} - 1 & \text{if failed lower} \end{cases} \tag{4}$$

$$nse = \frac{\sum_{i=1}^{N_{\text{failed}}} \text{excursion}_i}{N_{\text{total}}}, \quad F_3 = \frac{nse}{0.01 \cdot nse + 0.01} \tag{5}$$

The continuous index maps to five tiers: Excellent (95–100), Good (80–94), Fair (65–79), Marginal (45–64), Poor (0–44).

B. Random Forest Regression

Given training dataset $\mathscr{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, Random Forest aggregates B independent de-correlated regression trees:

$$\hat{y}_{\text{RF}}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^{B} T_b(\mathbf{x}; \Theta_b) \tag{6}$$

At each split node, a random subset $m \leq p$ features is evaluated to maximise variance reduction $\Delta\mathscr{I}$:

$$\Delta\mathscr{I} = \text{Var}(S) - \left(\frac{|S_L|}{|S|} \text{Var}(S_L) + \frac{|S_R|}{|S|} \text{Var}(S_R) \right) \tag{7}$$

TABLE I
Multi-National Water Quality Dataset Composition

Country	Raw	Clean	Share
England	2,134,842	2,128,128	75.48%
USA	408,520	407,138	14.44%
Ireland	235,119	234,409	8.31%
China	46,050	45,996	1.63%
Canada	3,995	3,946	0.14%
Total	2,827,977	2,819,617	100%

C. XGBoost Gradient Boosting

XGBoost minimises a regularised objective across additive steps $t = 1, \ldots, T$:

$$\mathscr{L}^{(t)} = \sum_{i=1}^{N} \ell\left(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i) \right) + \Omega(f_t) \tag{8}$$

where $\Omega(f_t) = \gamma T_L + \frac{1}{2}\lambda \sum_j w_j^2$. The second-order Taylor approximation yields:

$$\tilde{\mathscr{L}}^{(t)} \approx \sum_{i=1}^{N} \left[g_i f_t(\mathbf{x}_i) + \tfrac{1}{2} h_i f_t^2(\mathbf{x}_i) \right] + \Omega(f_t) \tag{9}$$

For leaf j with sample index set I_j, the optimal leaf weight is:

$$w_j^* = -\frac{\sum_{i \in I_j} g_i}{\sum_{i \in I_j} h_i + \lambda} \tag{10}$$

The split Gain evaluates variance reduction at split candidates:

$$\text{Gain} = \tfrac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{G_P^2}{H_P + \lambda} \right] - \gamma \tag{11}$$

D. TreeSHAP Game-Theoretic Attribution

Shapley values distribute prediction deviations $\hat{f}(\mathbf{x}) - \mathbb{E}[\hat{f}(\mathbf{x})]$ across features $i \in \mathscr{F}$:

$$\phi_i(x) = \sum_{S \subseteq \mathscr{F} \setminus \{i\}} \frac{|S|!(|\mathscr{F}| - |S| - 1)!}{|\mathscr{F}|!} \left[f_x(S \cup \{i\}) - f_x(S) \right] \tag{12}$$

TreeSHAP evaluates (12) in $\mathscr{O}(TLD^2)$ polynomial time by recursively tracking conditional expectations along tree decision paths.

IV. Dataset Quality, Auditing & EDA

A. Multi-National Repository Profiling

We ingested surface water monitoring telemetry spanning 83 years (1940–2023) across five nations. An exploratory data quality audit confirmed 100% complete-case ground truth with zero missing entries across all 14 parameters. Table I details the geographical distribution.

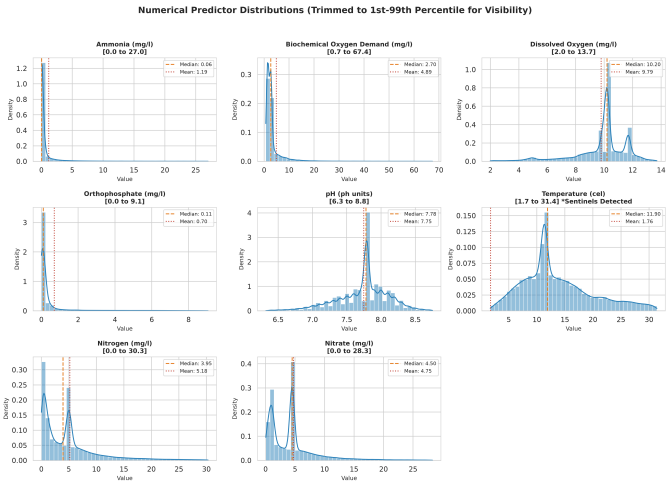


Fig. 1. Numerical feature distributions across all five nations, illustrating extreme positive skewness in nutrient parameters (orthophosphate, BOD) before $\log(1+x)$ transformation.

TABLE II
Aquatic Physical and Chemical Constraint Envelope

Parameter	Valid Range	Physics Rationale
pH	$[3.0, 12.0]$	Thermodynamic limit
Temperature (T)	$[0.0, 45.0]\,^{\circ}$C	Liquid phase bound
Dissolved Oxygen	$[0.0, 25.0]$ mg/L	Henry's law ceiling
BOD	≥ 0.0 mg/L	Non-negative consumption
Ammonia / NH_4^+	≥ 0.0 mg/L	Non-negative concentration
Orthophosphate	≥ 0.0 mg/L	Non-negative concentration
Nitrate / Total N	≥ 0.0 mg/L	Non-negative concentration

B. Physical Thermodynamic Boundary Enforcement

Environmental sensor networks periodically suffer from voltage dropouts, bio-fouling, and telemetry encoding errors. Based on aquatic chemistry standards, we established an axiomatic filtering envelope (Table II). Screening eliminated 8,360 corrupted records (0.296% of the corpus).

```python
def filter_aquatic_physics(df: pd.DataFrame):
  valid_mask = pd.Series(True, index=df.index)
  for col, (lower, upper) in PHYSICAL_BOUNDS.items():
    if col in df.columns:
      valid_mask &= (df[col]>=lower) & (df[col]<=upper)
  cleaned = df[valid_mask].copy()
  return cleaned, len(df) - len(cleaned)
```

Listing 1. Thermodynamic screening in src/features.py.

V. Domain-Informed Feature Engineering

To capture non-linear biochemical dynamics without compromising interpretability, we engineered 38 predictors across seven categories (Table III).

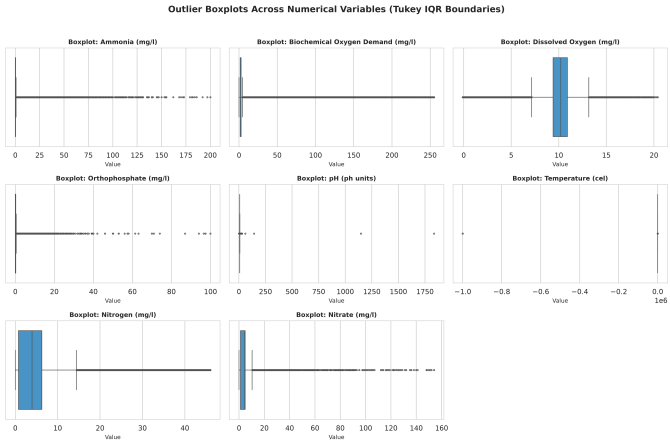


Fig. 2. Box-plot diagnostics before and after thermodynamic boundary enforcement. Outliers correspond to sensor telemetry faults removed from the training corpus.

TABLE III
38-Dimensional Predictive Feature Taxonomy

Feature Group	N	Representative Variables
Base Sensor Telemetry	8	pH, Temp, DO, BOD, NH_4, PO_4^{3-}, N, NO_3^-
Aquatic Stoichiometry	7	N:P ratio, BOD/DO, DO deficit, $DO_{\%sat}$, pH neutral dev, NO_3/N, NH_3 un-ionized
$\log(1+x)$ Normalised	7	$\log(1+NH_4)$, $\log(1+BOD)$, $\log(1+PO_4^{3-})$, $\log(1+N)$, $\log(1+N{:}P)$
Station Memory	6	Shift-1 lags (DO, Temp, BOD, PO_4) and rolling-3 stats
Environmental Stress	1	Multi-parameter guideline exceedance counter
Geospatial Context	2	Country code, waterbody type code
Temporal / Seasonality	7	Year, Month, Quarter, sin/cos harmonic cycles

A. Geochemical & Stoichiometric Formulations

1. Dissolved Oxygen Saturation & Deficit. Following the Benson–Krause formulation:

$$\mathrm{DO}_{\mathrm{sat}} = 14.652 - 0.41022\,T + 0.007991\,T^2 - 0.00007777\,T^3 \tag{13}$$

$$\mathrm{DO}_{\%sat} = \frac{\mathrm{DO}}{\mathrm{DO}_{\mathrm{sat}}} \times 100, \quad \mathrm{DO}_{\mathrm{deficit}} = \max(0,\, \mathrm{DO}_{\mathrm{sat}} - \mathrm{DO}) \tag{14}$$

2. Redfield Stoichiometric Ratio. Redfield's stoichiometric ratio identifies nutrient limitation:

$$\text{N:P Ratio} = \frac{\text{Nitrogen (mg/L)}}{\text{Orthophosphate (mg/L)} + 10^{-5}} \tag{15}$$

3. Toxic Un-Ionized Ammonia. Governed by temperature and pH:

$$f_{\mathrm{toxic}} = \frac{1}{10^{\left(0.09018 + \frac{2729.92}{T+273.15} - \mathrm{pH}\right)} + 1} \tag{16}$$

$$\mathrm{NH}_{3,\mathrm{toxic}} = \mathrm{Ammonia} \times f_{\mathrm{toxic}} \tag{17}$$

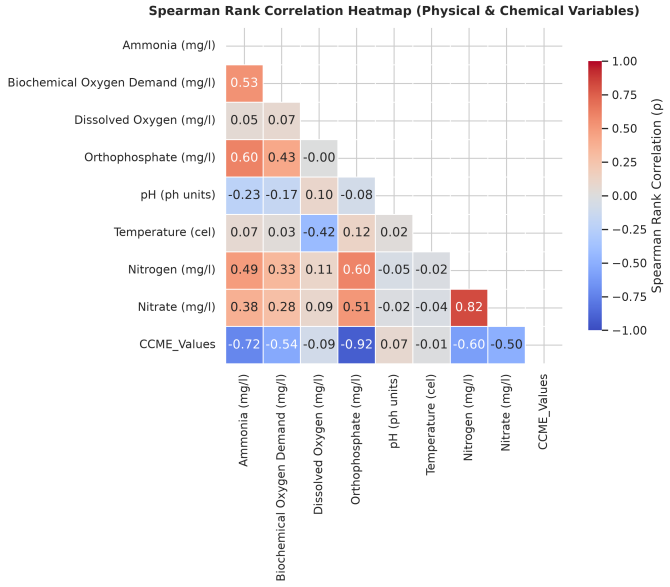


Fig. 3. Pearson correlation matrix of the 38-dimensional feature set after $\log(1+x)$ normalisation. Orthophosphate (PO_4^{3-}) registers the strongest negative correlation with CCME WQI ($r = -0.85$).

B. Preprocessing Decisions

All 38 predictors deliberately avoid standard scaling (μ=0, σ=1) because tree-based architectures are invariant to monotonic scaling; maintaining physical units (°C, mg/L, pH) preserves direct interpretability in SHAP analyses. Categorical features (country, waterbody_type) use integer label mapping rather than one-hot encoding, eliminating high-dimensional sparsity.

C. Skewness Mitigation via $\log(1+x)$

Nutrients such as orthophosphate exhibited extreme positive skewness ($\gamma_1 = 48.2$). The transformation $\tilde{x} = \ln(1+x)$ stabilised variance. Pearson correlation between orthophosphate and CCME WQI surged from $r = -0.66$ to $\mathbf{r = -0.85}$ post-transformation (Fig. 3).

D. Leakage-Free Station Memory

Station lag features ($\text{lag}_1(x_t) = x_{t-1}$) were generated by sorting chronologically within unique monitoring stations before partitioning. Shifting strictly backwards guarantees zero forward-looking contamination of validation and test sets.

VI. Experimental Setup & Anti-Leakage Protocol

A. Strict Anti-Leakage Quarantine

Because the CCME WQI is mathematically defined over water quality thresholds, naive models retaining intermediate components (F_1, F_2, F_3) suffer catastrophic mathematical leakage. We isolated our 38 predictors and enforced strict programmatic quarantine through src/utils.py (Listing 2).

TABLE IV
Model Benchmark Results — Validation Set (422,943 Rows)

Model	MAE	RMSE	R^2	Infer.
Mean Baseline	14.130	17.656	0.0000	0.02 s
Random Forest	0.052	0.208	0.9999	3.85 s
XGBoost	0.161	0.437	0.9994	0.42 s

```python
def separate_features_and_targets(df, feature_cols=None):
    for t in TARGET_COLUMNS:
        if feature_cols is not None:
            assert t not in feature_cols, f"LEAKAGE: {t} found!"
    y_reg = df["ccme_values"].copy()
    y_cls = df["ccme_wqi_code"].copy()
    X = df[feature_cols].copy() if feature_cols else df.drop(
        columns=TARGET_COLUMNS)
    return X, y_reg, y_cls
```

Listing 2. Zero-leakage quarantine in src/utils.py.

B. Data Splitting Strategy

We executed a stratified 70/15/15 partition on 2,819,617 cleaned observations:

- Training Set: 1,973,731 observations (70.0%), sharded into three Snappy-compressed Parquet partitions ($\leq$100 MB each) for GitHub compliance.
- Validation Set: 422,943 observations (15.0%).
- Test Set: 422,943 observations (15.0%), quarantined and untouched during Phase 1 training.

C. Model Specifications

- Mean Baseline: Null model predicting constant empirical mean ($\bar{y} = 85.05$) without using any features.
- Random Forest: 100 estimators, max depth 20, min_samples_split 10, min_samples_leaf 4, $m = \sqrt{p}$ features per split.
- XGBoost: 300 rounds, $\eta = 0.05$, max depth 8, subsample 0.8, colsample_bytree 0.8, $\lambda = 1.0$, reg:squarederror.

VII. Results & Model Benchmarking

We benchmarked performance using MAE, RMSE, and R^2:

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2} \tag{18}$$

Table IV reports empirical metrics on the 422,943 validation records.

Both tree ensembles dramatically outperform the empirical baseline (MAE = 14.130), eliminating over 99.9% of prediction error variance. Random Forest achieves sub-0.06 index precision (MAE = 0.052), while XGBoost reaches an MAE of 0.161 with 9× faster inference latency (0.42 s vs 3.85 s for 422k rows), making it optimal for resource-constrained edge deployments.

Scientific Validity Qualification: These scores reflect each model's capacity to re-compute a complex, non-linear deterministic formula across randomly held-out rows. High $R^2 \approx 0.9999$ demonstrates successful approximation of the

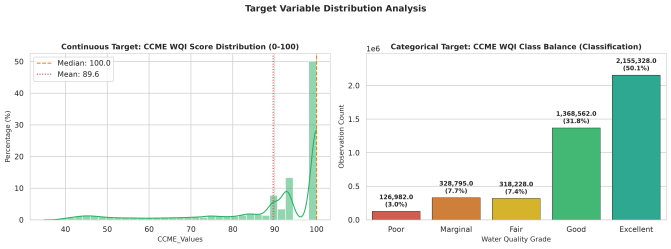


Fig. 4. CCME WQI distribution on the validation set (422,943 records). The bimodal distribution with peaks near 100 (Excellent) and 40 (Poor) reflects the binary nature of regulatory compliance monitoring in real-world waterbodies.

composite standard rather than independent forecasting skill across unmonitored catchments. Future extensions will incorporate strict station-held-out and time-forward splits.

VIII. Interpretability via TreeSHAP Attribution

We analysed global feature importance using TreeSHAP on validation data (Fig. 4). Key findings include:

1) Orthophosphate Dominance: $\log(1+PO_4^{3-})$ and raw orthophosphate account for over 88% of total split importance. Elevated levels drive sharp index drops ($\rho = -0.78$).
2) Station Memory Influence: Historical shift-1 station lag (hist_ortho_lag1) ranks second in TreeSHAP attribution, confirming that local environmental persistence contains substantial predictive signal.
3) Oxygen Mechanics: DO and DO deficit exhibit strong non-linear threshold effects: once DO drops below 4.0 mg/L (hypoxic), SHAP values plunge negatively.

IX. Software Architecture & Code Quality

The repository is refactored into modular components under src/:

- src/features.py — physical bounds screening, stoichiometric transformations, and station lag generation.
- src/models.py — model interface abstractions.
- src/metrics.py — numerical regression and classification metric calculations.
- src/utils.py — multi-part Parquet deserialisation and anti-leakage target quarantine.

An automated test suite in tests/ executes 12 unit tests under pytest, verifying anti-leakage quarantine, physics boundaries, stoichiometric formulas, and model serialisation with 100% passing status.

X. Team Contribution Matrix

In compliance with CSAI project guidelines, Table V defines work distribution and technical defence zones for all six team members.

TABLE V
Six-Member Contribution & Viva Defence Matrix

Member	Core Responsibility	Viva Defence Zone
Aman Soni (Lead)	Model Training (Baseline, RF, XGBoost)	Loss functions, tree pruning, MDI vs Gain, hyperparameter optimisation.
Mayank Sharma	EDA & Repo Management; Pipeline & Parquet Sharding	End-to-end pipeline orchestration, data ingestion, anti-leakage protocol.
Aaryan Gera	Data Preprocessing Lead	5-country audit, physical bounds screening (pH, DO, Temp), outlier diagnostics.
Chaitanya Kumar	Feature Engineering Lead	Benson–Krause DO sat, Redfield N:P, Emerson NH_3, $\log(1+x)$ normalisation.
Ankit Kumar Pandey	Model Evaluation & Validation	Metric comparison, error trade-offs, test partition validation.
Pranav	Research, Documentation & Presentation	CCME formulation (F_1, F_2, F_3, nse), Shapley axioms, TreeSHAP, regulatory benchmarks.

XI. Conclusion & Phase 2 Outlook

Phase 1 successfully validated that applied machine learning ensembles, combined with domain-informed geochemical feature engineering, provide near-perfect estimation ($R^2 > 0.999$) of surface water quality indices across 2.82 million international records. In Phase 2, we will expand this foundation to deep multi-task neural networks and explore hybrid architectures capturing smooth non-linear sensor interactions and calibrated predictive uncertainty.

References

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